Edward Silva

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Experience

Embedded Software Engineer, SpaceX — Jul 2026 – Present
C, C++, Python, Git — Bastrop, TX

- Write embedded software in C and C++ for Starlink phased array systems
- Improved the run time of math used in the Starlink phased array software

Software Engineering Intern, Kratos Defense — Jun – Aug 2025
C++, Python, SIMD, Git — Colorado Springs, CO

- Sped up legacy real-time DSP code in C++ by 1.6x through refactoring, AVX-512 vectorization, and Intel VTune profiling on Linux over SSH
- Presented a faster FIR filter implementation and its performance trade-offs to the team for use on later projects
- Built a configurable logging framework with severity levels and error tracking to cut debug time and improve maintainability of the real-time DSP software

Co-op Intern, Electrical Design, Jordan and Skala Engineers — Jan – Jun 2025
Autodesk Revit, AutoCAD MEP, NEC Compliance — Denver, CO

- Performed circuit loading and voltage drop calculations, sized breakers to NEC requirements, and drafted electrical layouts and design documentation for 20+ multi-unit residential and non-residential projects

Undergraduate Researcher, ePower Hubs Research Lab — Aug – Dec 2024
Golden, CO

- Reviewed the literature on sensor systems and farm-level control strategies for offshore wind, under variable voltage, power, and frequency constraints
- Wrote the findings up in internal LaTeX reports on cost reduction in wind farm maintenance, design, and grid integration

Projects

Electromagnetic Bicycle Damper (EMBR), MATLAB, Python — Fall 2025 – Spring 2026

- Owned the electromagnet design and calculation verification within an 8-person interdisciplinary team, sponsored by ARUS MR Tech, that retrofitted a production front suspension fork with a semi-active MR damper
- Modeled magnetic field strength and spatial geometry using the Biot-Savart law and complete elliptic integrals in Python, ensuring field lines oriented perpendicularly to fluid flow to maximize yield stress activation
- Confirmed in testing that the electromagnet worked as designed and produced a physically noticeable change in damping

Non-Linear State Estimation for the EMBR Damper, MATLAB, Simulink — Spring 2026

- Designed a non-linear state estimation strategy for the semi-active EMBR suspension, modeling the interactions of a custom magnetorheological (MR) fluid damper, MOSFET power regulation, and a multi-sensor array
- Derived continuous and discrete-time non-linear state-space models in MATLAB and Simulink, simulating system kinematics to validate damping performance against harmonic, stochastic, and discrete terrain disturbances
- Implemented and compared Extended (EKF) and Unscented (UKF) Kalman Filters to estimate unmeasured states

Pitch-Tracking Systems in Auto-Tune, MATLAB — Spring 2026

- Measured latency and residual error to verify the performance of real-time Fast Fourier Transform (FFT) and Autocorrelation (ACF) pitch-tracking algorithms in MATLAB
- Designed automated data visualization scripts to quantify computational processing time, demonstrating a 2.68x execution speedup for FFT over ACF

Gantry Crane Control System, MATLAB, Simulink Fall 2025

- Designed a 50 Hz discrete-time controller for a gantry-crane system by linearizing the plant model, constraining cart velocity to <= 1 m/s and achieving 0% overshoot for 1 m position step commands
- Verified system robustness through frequency-domain analysis, proving stability margins including a 48° phase margin and a ∼0.5 s delay tolerance
- Implemented an LQR state-space controller featuring a full-order observer and anti-windup tracking; evaluated reference-scaling against integral-augmented topologies to achieve a 3.45 s settling time with negligible overshoot and near-zero steady-state error under saturation

Autonomous Maze-Navigating Robot, Python Fall 2025

- Achieved the second-fastest course completion time in the cohort with an autonomous maze-navigating robot built by a 4-person team, integrating computer vision, embedded control, and I2C communication
- Built the computer vision subsystem with one teammate, using Python and OpenCV with ArUco marker detection to extract real-time distance and angle metrics for navigation
- Interfaced the vision subsystem with motor control via I2C bus, transmitting telemetry data to an Arduino while updating a real-time RGB LCD display

Hydraulic Ram Pump Fall 2022

- Built a working hydraulic ram pump prototype for under $100 within a 5-person team, using PVC piping and a pH filtration system to mitigate acid mine drainage at legacy mine sites
- Received the Subject Matter Expert Award at the final presentation

Skills

Skills: C++, Python, C, SIMD, MATLAB, Simulink, Git, Arduino

Education

Colorado School of Mines May 2026

BS Electrical Engineering, Minor in Computer Science, Golden, CO

Honors: Dean's List (Fall 2022, Fall 2024, Spring 2026) | Honor Roll (Fall 2023, Spring 2024, Fall 2025)

Scholarships: Provost Scholarship (Merit scholarship) | 2023 - 2024 C-MAPP Scholar (Frontier Technology Inc.) | 2024 - 2025 C-MAPP Scholar (Parsons Corporation) | 2024 - 2025 American Bureau of Shipping Scholar (Merit scholarship)

Electrical Engineering Courses: Digital Signal Processing, Estimation Theory & Kalman Filtering, Modern Control Design, Embedded Systems, Field Session – Printed Circuit Board Design, Electrical Engineering Integration and Design, Introduction to Feedback Control Systems, Field Session – Computational Methods for Electrical Engineers, Electronic Devices and Circuits, Fundamentals of Engineering Electromagnetics, Electrical Circuits Laboratory, Digital Logic, Information Systems Science I, Information Systems Science II, Fundamentals of Electric Machinery

Computer Science Courses: Algorithms, Computer Organization, Software Engineering, Data Structures and Algorithms, Elements of Computing Systems, Discrete Mathematics, Introduction to the Linux Operating System, Programming Challenges I

Math and Design Courses: Linear Algebra, Differential Equations, Capstone Design I, Capstone Design II

Certifications

MATLAB Machine Learning Techniques, MathWorks
MTA: Introduction to Programming Using Java, Microsoft
MTA: Introduction to Programming Using Python, Microsoft

Organizations

IEEE-HKN, Nu Pi Chapter, Colorado School of Mines, Member 2026 – Present
IEEE, Member 2025 – Present
IEEE Young Professionals, Member 2025 – Present
Denver Astronomical Society, Member 2022 – 2025